PARISHRAM 2.0 2026 (Class 12th)

CHEMISTRY

DPP 4

d & f-Block Elements

Q1 The major product formed in the given process is;

$$MnO_4^{2-} \xrightarrow[\text{in alkaline solutuion}]{\text{electrolytic oxidation}}$$

(A) MnO_2 (B) MnO_4^-

(C) Mn_2O_7 (D) MnO

Q2 Potassium permanganate on heating at $513\ K$ gives a product which is:

(A) Paramagnetic and colourless

(B) Diamagnetic and green

(C) Diamagnetic and colourless

(D) Paramagnetic and green

Q3 The product of the following reaction will be : $I^- + MnO_4^- \rightarrow$

(A) $IO_3^{\ominus}$

(B) $IO^{\ominus}$

(C) $IO_4^{\ominus}$

(D) $I_3^{\ominus}$

Q4 Pyrolusite is MnO_2 used to prepare $KMnO_4$. Steps are,

$$MnO_2 \xrightarrow{I} MnO_4^{2-} \xrightarrow{II} MnO_4^-$$

Steps I and II are respectively :

(A) Fuse with KOH/air and electrolytic oxidation.

(B) Fuse with KOH/KNO_3 and electrolytic oxidation.

(C) Both (a) and (b).

(D) Dissolve in H_2O and oxidation.

Q5 The reaction of aqueous $KMnO_4$ with H_2O_2 in acidic conditions gives:

(A) Mn^{4+} and O_2

(B) Mn^{2+} and O_2

(C) Mn^{2+} and O_3

(D) Mn^{4+} and MnO_2

Q6 Potassium manganate (K_2MnO_4) is formed when:

(A) Cl_2 is passed into an aqueous solution of $KMnO_4$

(B) MnO_2 is fused with KOH in air

(C) Formaldehyde reacts with $KMnO_4$ in presence of strong alkali

(D) $KMnO_4$ reacts with concentrated H_2SO_4

Q7 Which of the following compounds has colour but no unpaired electrons?

(A) $KMnO_4$

(B) K_2MnO_4

(C) $MnSO_4$

(D) $MnCl_2$

Answer Key

Q1	B
Q2	D
Q3	A
Q4	A
Q5	B
Q6	B
Q7	A

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 **Text Solution:**

$$MnO_4^{2-} \xrightarrow[\text{in alkaline solution}]{\text{electrolyic oxidation}} MnO_4^-$$

Video Solution:

Q2 **Text Solution:**

$\mathrm{KMnO_4} \xrightarrow{\Delta} \mathrm{K_2MnO_4 + MnO_2 + O_2}$

$\mathrm{K_2\,MnO_4}$ is green is colour and paramagnetic.

Video Solution:

Q3 **Text Solution:**

$\mathrm{IO_3^{\ominus}}$

Video Solution:

Q4 **Text Solution:**

Fuse with $\mathrm{KOH}/$ air and electrolytic oxidation.

Video Solution:

Q5 **Text Solution:**

$\mathrm{Mn^{2+}}$ and $\mathrm{O_2}$

Video Solution:

Q6 **Text Solution:**

$$\mathrm{2\,MnO_2 + 4\,KOH + O_2 \rightarrow 2K_2\,MnO_4 + 2\,H_2O}$$

Video Solution:

Q7 **Text Solution:**

$\mathrm{KMnO_4}$ has colour but no unpaired electrons.

Video Solution: